

PREMENSTRUAL TENSION DISORDER



Their well being
depends on our
understanding



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OBJECTIVES



Understand PMS: Define PMS, its significance, and clinical features.

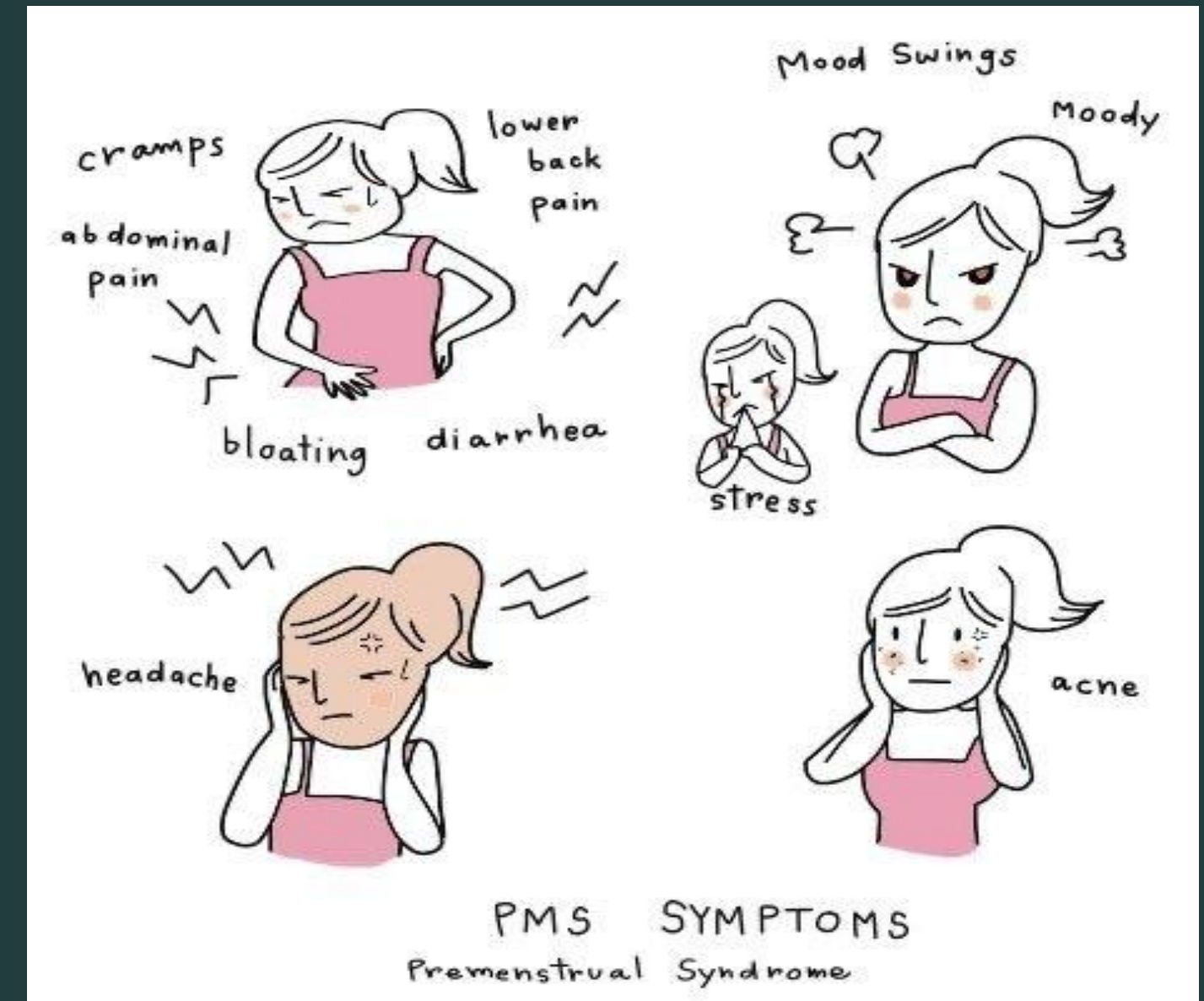


Explain Pathophysiology and Diagnosis: Describe underlying mechanisms and diagnostic criteria.



Approach and Management: Outline clinical evaluation and evidence-based management strategies.

Premenstrual syndrome (PMS) is a common disorder in women of reproductive age. It is characterized by at least one physical, emotional, or behavioral symptom, which appears in the luteal phase of the menstrual cycle and resolves shortly after the onset of menses.

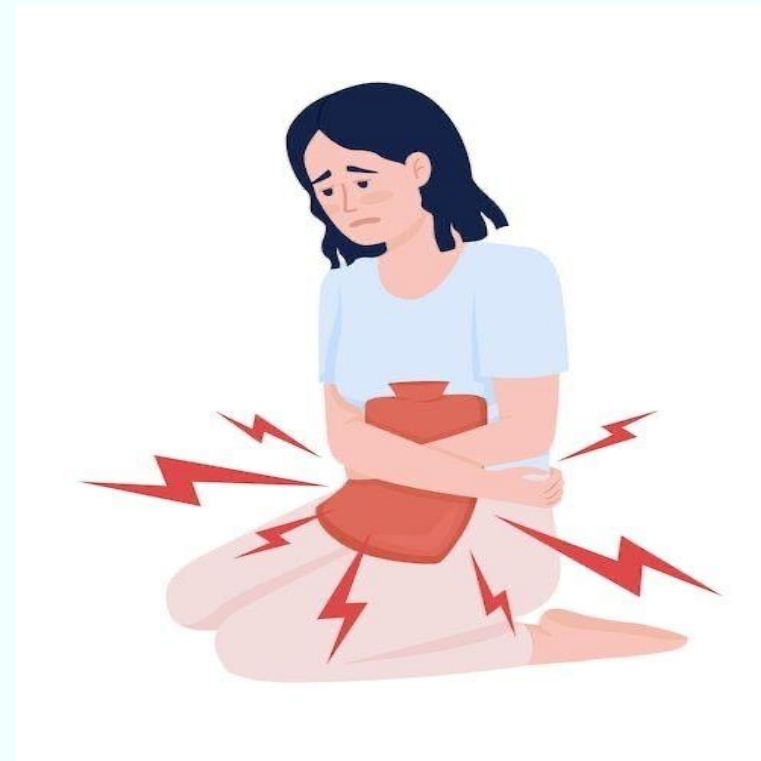


EPIDEMIOLOGY

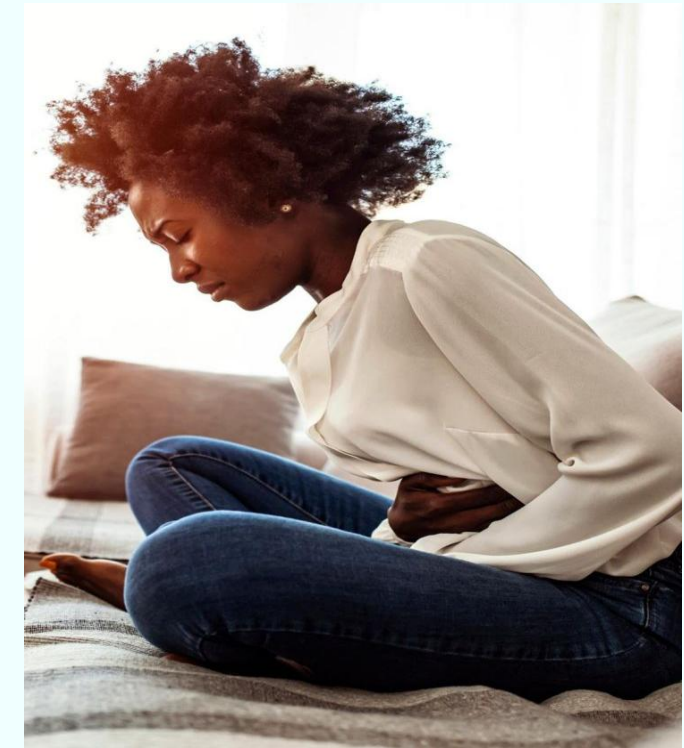
Epidemiological studies have revealed that about 80% to 90% of women manifest at least one of the PMS signs; however, in about 2.5% to 3% of women, the syndrome is severe enough to affect their activities and social communications.



- **Most common in women aged 25–40.**



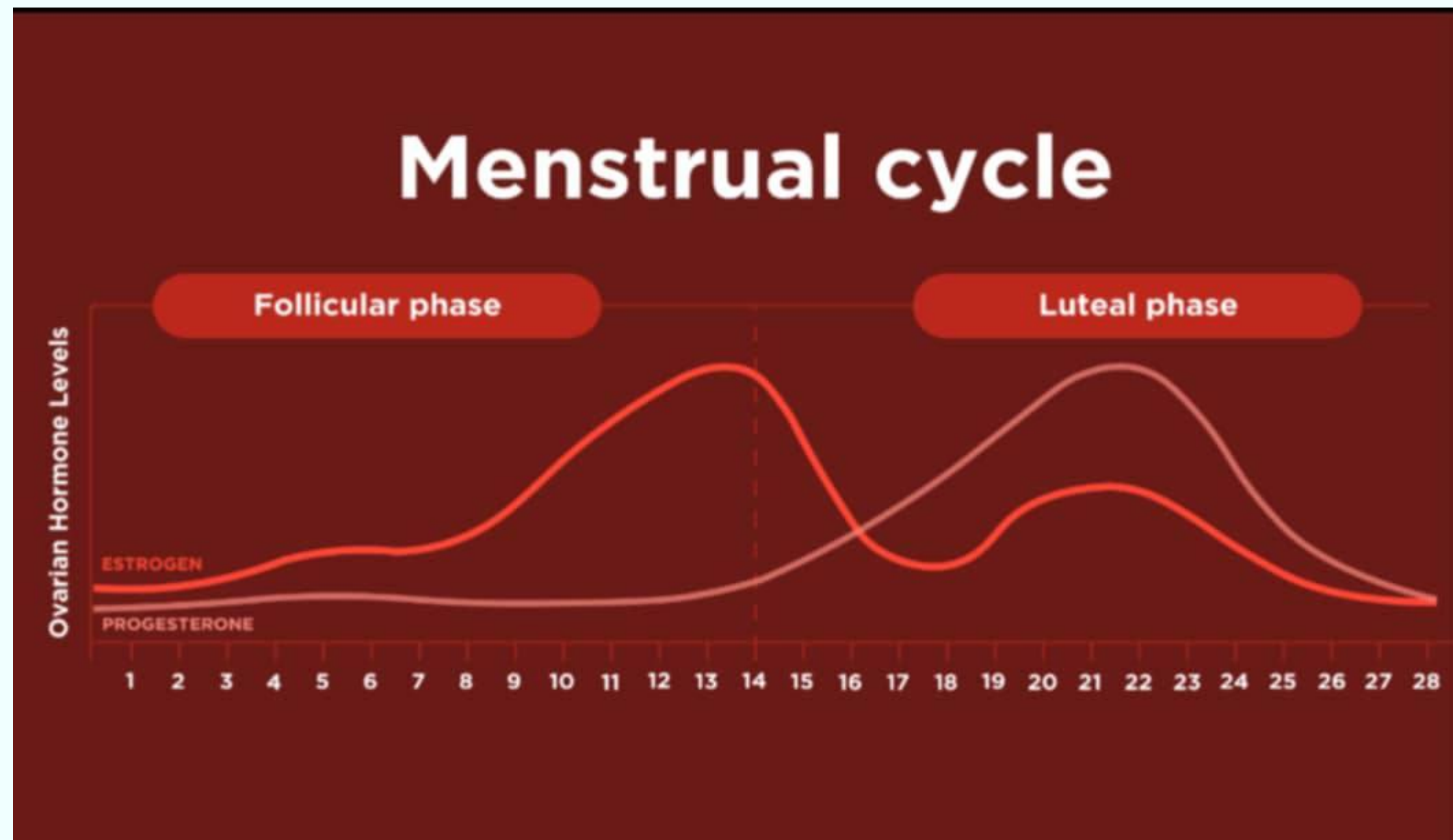
- **More frequent in women with existing mood disorders.**



- **Shows a familial/genetic tendency.**

PATHOPHYSIOLOGY

One of the proposed etiologies of PMS involves the cyclic hormonal fluctuations of estrogen and progesterone during the menstrual cycle, particularly during the late luteal phase as the body prepares for menstruation. Women with and without PMS have similar cyclic changes in ovarian steroid concentrations. The distinction between having or not having PMS could stem from how the body responds to the cyclic changes in ovarian hormones across the menstrual cycle.



PATHOPHYSIOLOGY

Role of neurotransmitters:

Serotonin: During the luteal phase, women with PMS have more serotonin abnormalities than women without PMS symptoms. Serotonin is responsible for many psychological, physiologic, emotional, and behavioral responses and women with PMD express similar symptoms to individuals who have a reduced level of serotonin, posing the thought that the two could be related.

gamma aminobutyric acid (GABA): present work suggests that progesterone may play a modulatory role in GABA's inhibitory system. This hypothesis is supported by observations of improvement in PMS symptoms using agents such as benzodiazepine and alprazolam that work to increase GABA activity. In addition, metabolites of progesterone have been shown to function as positive allosteric modulators of GABA in the brain.

Endorphins: evidence that shows the release of endorphins is coupled with the release of estrogen and could theoretically decline with ovulation and remain at low levels during the late luteal phase when estrogen levels decline again prior to menstruation.

SYMPTOMS

physical symptoms :

Abdominal cramps, abdominal bloating, tenderness and pain in the breasts, pain (back, head, muscle, joints), swelling of hands and feet, weight fluctuations, constipation, and diarrhea.

Psycho-emotional symptoms:

mood swings, Depression, anxiety, tendency to cry, irritability, emotional lability, difficulty, concentrating, and anger.

Behavioral symptoms:

changes in libido and sexual desire, social withdrawal from partner, family and friends, sensitivity, increased appetite, decreased interest in activities, and restlessness.

RISK FACTORS FOR PREMENSTRUAL SYNDROME (PMS)

Family history of PMS or mood disorders, suggesting a genetic or hereditary component.

Psychiatric comorbidities, including a history of depression or anxiety, increase susceptibility to premenstrual symptom exacerbation.

Stress and psychosocial factors, such as high perceived stress or early childhood adversities.

Lifestyle factors, including poor sleep, inadequate physical activity, and high caffeine intake.

Younger reproductive age, with symptoms often more prominent in women in their late teens to early 30s.

DIFFERENTIATION FROM PMDD

Premenstrual Syndrome (PMS) and Premenstrual Dysphoric Disorder (PMDD) represent distinct points on the spectrum of premenstrual symptomatology but differ markedly in severity, diagnostic requirements, and associated **functional impairment**.

PMDD is a formally recognized psychiatric disorder defined by **DSM-5 criteria**, requiring the presence of **five or more symptoms**, including **at least one core affective symptom** (e.g., pronounced irritability, marked affective lability, depressed mood, or significant anxiety/tension).

DIFFERENTIATION FROM OTHER DISORDERS

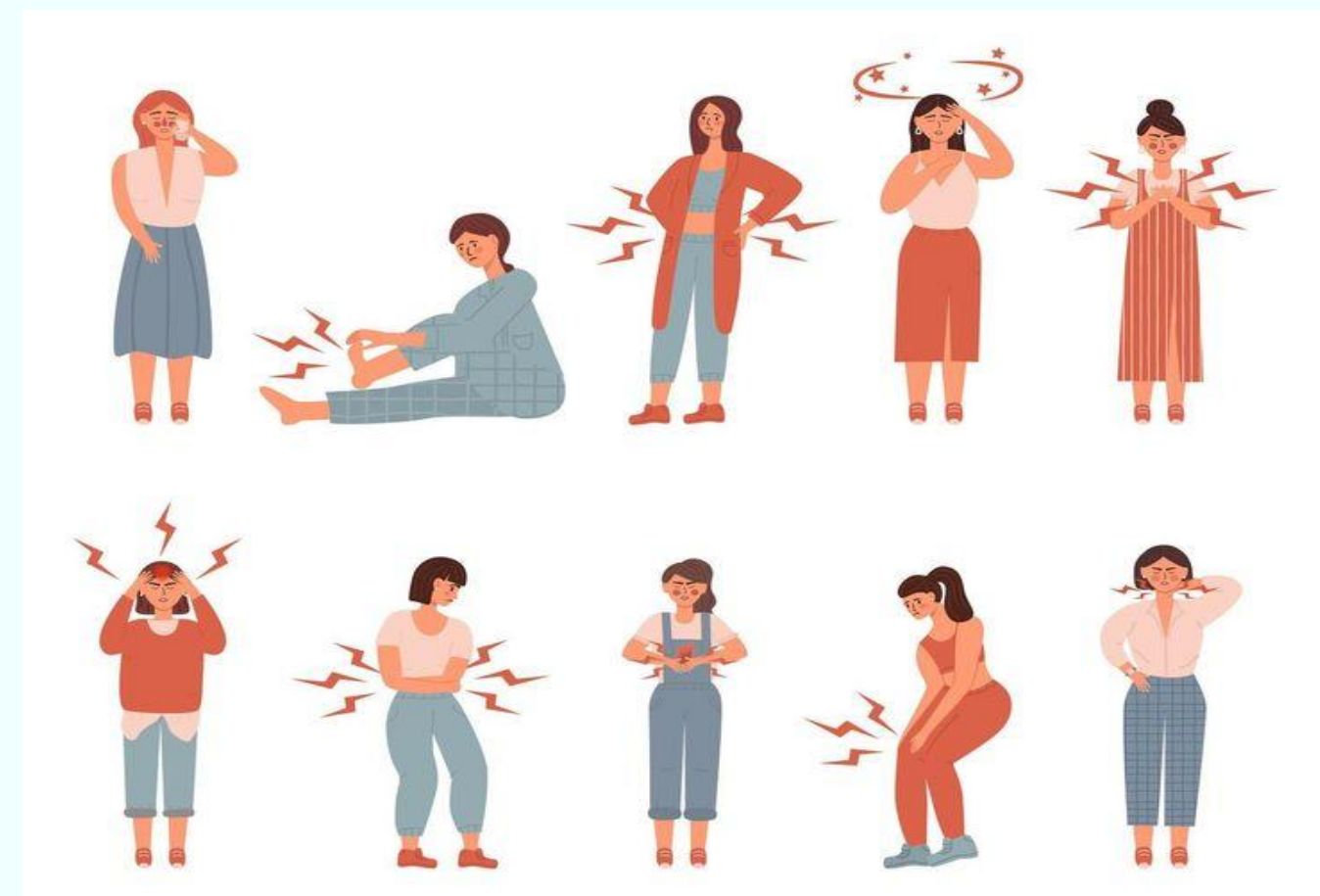
PMS is identified by its **strict luteal-phase cyclicity**, whereas other medical or psychiatric disorders present with **persistent or non-cyclic** symptoms. This temporal pattern distinguishes PMS from conditions such as major depressive disorder, bipolar disorder, hypothyroidism, and anemia.

Why does everything feel
like a plot twist today?!!!



DIAGNOSIS

The diagnosis of PMS is established clinically and requires that symptoms **occur in the luteal phase** and be confirmed over **a minimum of two consecutive cycles**. Symptoms must **resolve within four days of the onset of menstruation** and remain absent during the mid-follicular phase. Diagnostic criteria further stipulate the presence of **at least one affective and one somatic symptom**. These symptoms must **not be attributable to any other medical or psychiatric condition**, necessitating appropriate clinical exclusion.



INVESTIGATIONS FOR PREMENSTRUAL SYNDROME (PMS)

Prospective symptom charting for *a minimum of two cycles* is essential to confirm timing, recurrence, and severity.

Other investigations are performed to exclude alternative causes, including:

Thyroid function tests (to rule out hypothyroidism)

Complete blood count (to exclude anemia)

Screening for psychiatric disorders, particularly major depressive disorder and bipolar disorder

Differentiation from PMDD.



TREATMENT

Because the etiology of PMS and PMDD is not clear, symptom relief is the goal of treatment.

There are Different treatment modalities according to the severity of symptoms and patient preference, including:

- 1. Nonpharmacological**
- 2. Pharmacological**
- 3. Surgical**



NONPHARMACOLOGICAL TREATMENTS

Lifestyle changes include stress reduction, alcohol and caffeine limitation, and exercise, yoga and meditation. Cognitive-behavioral therapy (CBT) and other psychological counselling treatments should also be considered as an initial treatment option.

dietary calcium and vitamin D intake play a role in PMS symptoms. Women who had a high dietary calcium intake (1.2 g per day) were 30 percent less likely to have PMS symptoms compared with women who had a low intake (0.5 g). Women with high daily vitamin D intake were also less likely to have PMS symptoms.

Also, Supplementation with vitamin B6 decreases PMS symptoms by more than two folds.

Complementary and Alternative Therapies like ginkgo, saffron, St. John's wort, and soy. Currently, there is no sufficient evidence to support their effect in relieving PMS symptoms, however, regarding chastberry, one RCT shows that supplementation with 20 mg of chastberry daily can reduce more than 50% of PMS symptoms. Primrose oil is also effective.

Pharmacological treatment

The selective serotonin reuptake inhibitors (SSRIs) like citalopram (Celexa), escitalopram (Lexapro), fluoxetine (Prozac), and sertraline (Zoloft)—taken daily or only during the luteal phase of menstruation—significantly decrease the physical and psychological symptoms of PMS.

Note, paroxetine is regarded as pregnancy category D by FDA, as causes major congenital defects so should be avoided in a women in child bearing age without contraception or sterilization

Combined oral contraceptive pills COCP: the most effective regimen appears to be bicycling or tricycling pill packets (i.e, taking two or three packets in a row without a scheduled break).

transdermal oestrogen also has been shown to significantly reduce PMS symptoms by overcoming the fluctuations of the normal cycle.

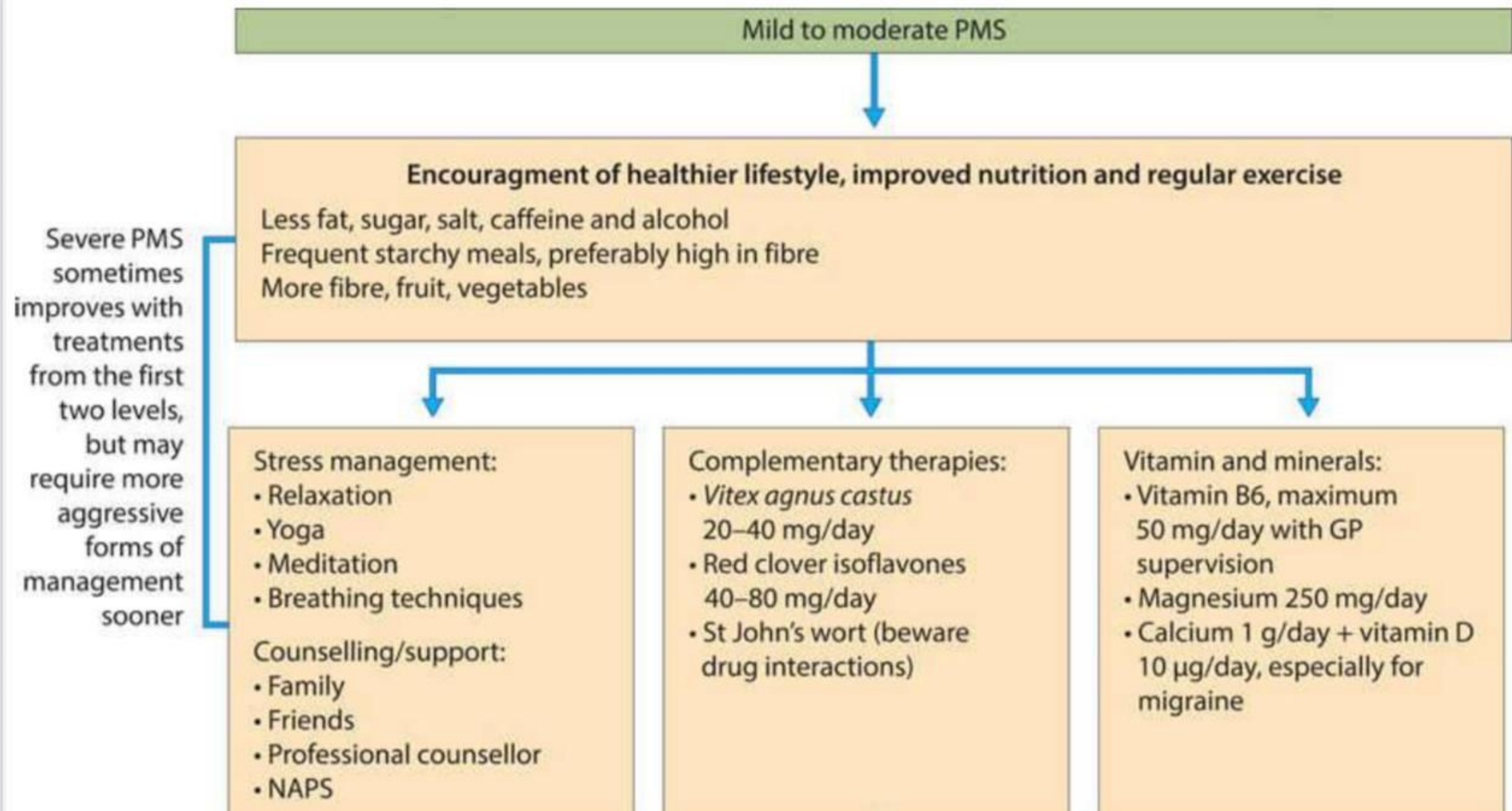
GnRH analogues eg, goserelin (Zoladex), histrelin (Vantas), are very effective in treatment for PMS as they turn off ovarian activity. To reduce the risk of osteoporosis it is recommended that a continuous combined form of hormone replacement therapy is administered concurrently(i.e, add back HRT).

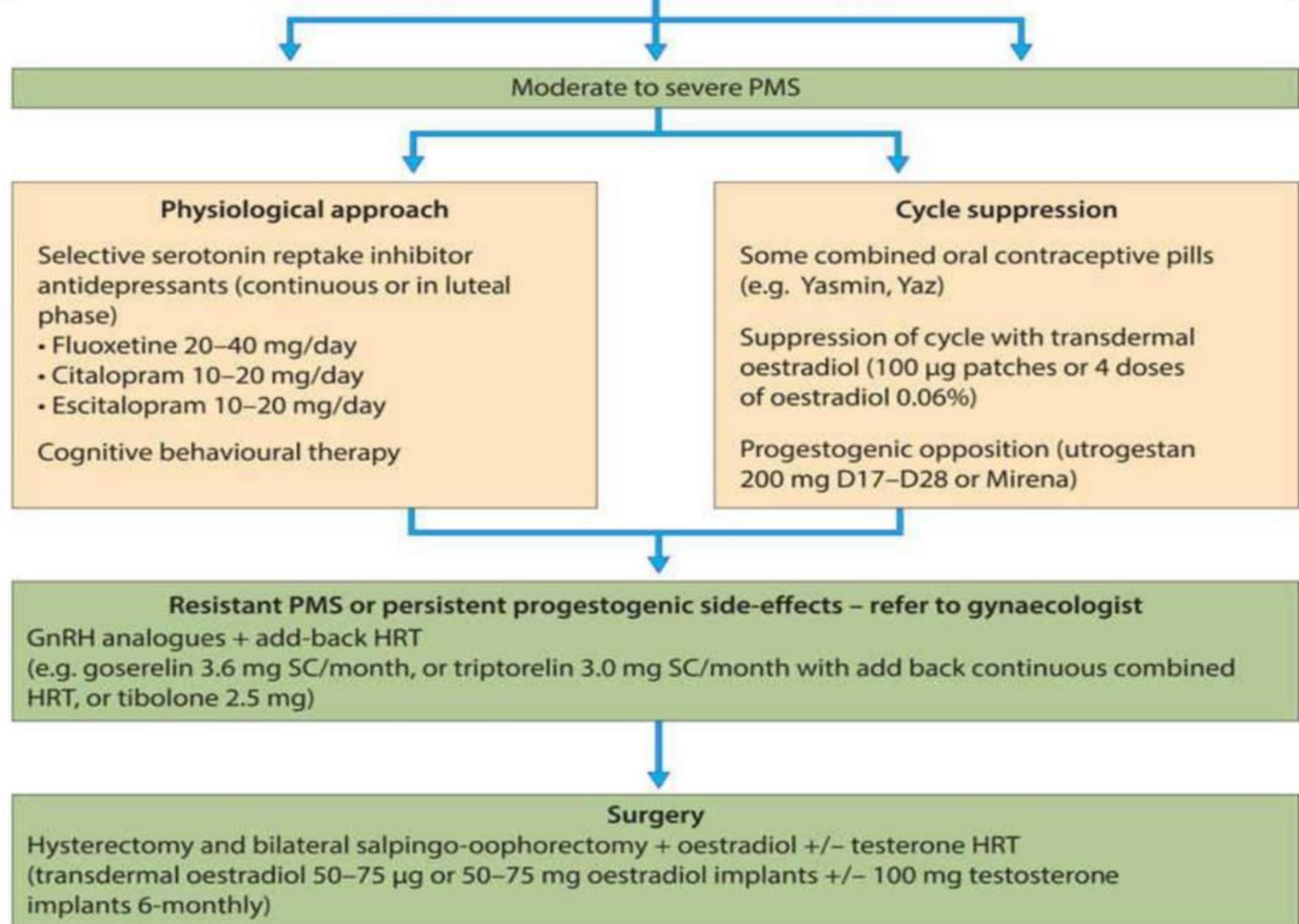
SURGICAL TREATMENT

Hysterectomy with bilateral salpingo-oophorectomy: This procedure obviously completely removes the ovarian cycle. It should only be performed if all other treatments have failed. It is essential for such patients to have a preoperative trial of GnRH analogue as a 'test' to ensure that switching off ovarian function (by removing the ovaries at hysterectomy) will indeed cure the problem



TREATMENT OVERVIEW





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THANKS!



*i suffer from p . m . s
..... pissed at mens syndrome*